Object Oriented Programming (CSEG1044)

Unit 06 – Lecture 05: Swing UI Panels + Integration

Tofik Ali

School of Computer Science, UPES Dehradun

February 27, 2026

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Plan screens/panels and choose layout managers.

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Learning Outcomes

- Plan screens/panels and choose layout managers.
- Implement events that call service/DAO cleanly.

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Learning Outcomes

- Plan screens/panels and choose layout managers.
- Implement events that call service/DAO cleanly.
- Complete Deliverable-4: one end-to-end UI flow connected to persistence.

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Where we are in the capstone

- Deliverable-0: team + repo + backlog

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Where we are in the capstone

- Deliverable-0: team + repo + backlog
- Deliverable-1: requirements + use cases

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Where we are in the capstone

- Deliverable-0: team + repo + backlog
- Deliverable-1: requirements + use cases
- Deliverable-2: class design + packages

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Where we are in the capstone

- Deliverable-0: team + repo + backlog
- Deliverable-1: requirements + use cases
- Deliverable-2: class design + packages
- Deliverable-3: DB schema + DAO CRUD

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- Deliverable-0: team + repo + backlog
- Deliverable-1: requirements + use cases
- Deliverable-2: class design + packages
- Deliverable-3: DB schema + DAO CRUD
- **Now: Deliverable-4** UI → DB → UI refresh

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Deliverable-4 (today)

- At least 2 screens/panels (list + form).

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Deliverable-4 (today)

- At least 2 screens/panels (list + form).
- One complete flow: Add (or Book) → save in DB → refresh list.

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Deliverable-4 (today)

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- One complete flow: Add (or Book) → save in DB → refresh list.
- Validations + user-friendly messages.

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Deliverable-4 (today)

- At least 2 screens/panels (list + form).
- One complete flow: Add (or Book) → save in DB → refresh list.
- Validations + user-friendly messages.
- Evidence: issues + PR + short README "how to run".

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Unit 06: Capstone Project

- Lecture focus: Swing UI Panels + Integration

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Today: Roadmap

- Screen plan + navigation (what panels do we need?)

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Today: Roadmap

- Screen plan + navigation (what panels do we need?)
- Layout managers + event handling pattern

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Today: Roadmap

- Screen plan + navigation (what panels do we need?)
- Layout managers + event handling pattern
- Clean integration: UI vs service vs DAO; validation + messages

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Screen plan example (Resource Booking)

- Resource list (table) + “Add Resource” (dialog/form)

Navigation choices:

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Screen plan example (Resource Booking)

- Resource list (table) + "Add Resource" (dialog/form)
- Booking form (select resource, time, user)

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Screen plan example (Resource Booking)

- Resource list (table) + "Add Resource" (dialog/form)
- Booking form (select resource, time, user)
- Booking list (table)

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- Resource list (table) + "Add Resource" (dialog/form)
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- Booking list (table)

Navigation choices:

- `CardLayout` to switch screens

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Screen plan example (Resource Booking)

- Resource list (table) + "Add Resource" (dialog/form)
- Booking form (select resource, time, user)
- Booking list (table)

Navigation choices:

- `CardLayout` to switch screens
- `JTabbedPane` for independent pages

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Layout managers (cheat sheet)

- `BorderLayout`: main structure (North/West/Center)

Rule: avoid `null` layout.

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Layout managers (cheat sheet)

- `BorderLayout`: main structure (North/West/Center)
- `BoxLayout`: vertical stacks

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Layout managers (cheat sheet)

- `BorderLayout`: main structure (North/West/Center)
- `BoxLayout`: vertical stacks
- `GridLayout`: uniform grids

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- `BorderLayout`: main structure (North/West/Center)
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- `GridBagLayout`: aligned forms (labels + fields)

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Layout managers (cheat sheet)

- `BorderLayout`: main structure (North/West/Center)
- `BoxLayout`: vertical stacks
- `GridLayout`: uniform grids
- `GridBagLayout`: aligned forms (labels + fields)
- `CardLayout`: multi-screen apps

Rule: avoid `null` layout.

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Event handler pattern (keep it short)

```
btnSave.addActionListener(e -> {
    try {
        service.createResource(
            txtName.getText(),
            txtType.getText(),
            Integer.parseInt(txtCap.getText())
        );
        JOptionPane.showMessageDialog(this, "Saved!");
        refreshTable();
    } catch (Exception ex) {
        showError(ex);
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```

- UI reads inputs, service validates, DAO persists.

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```

- UI reads inputs, service validates, DAO persists.
- UI shows message and refreshes view.

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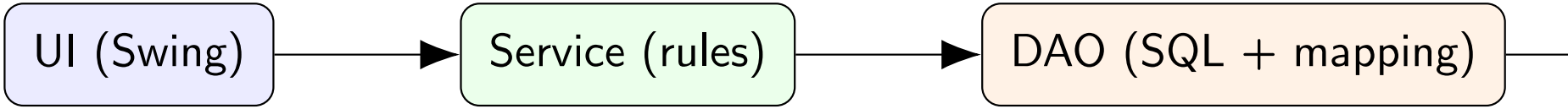
Layering (what goes where?)



- UI: *no SQL*; convert fields + show results.

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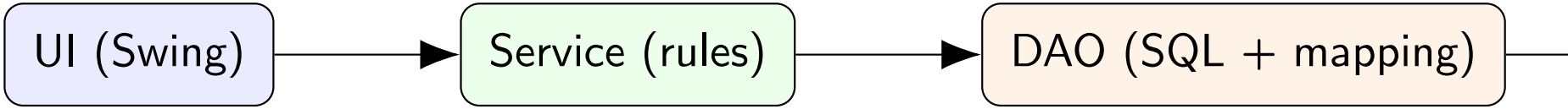

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- Service: validations; business rules (e.g., no overlap).

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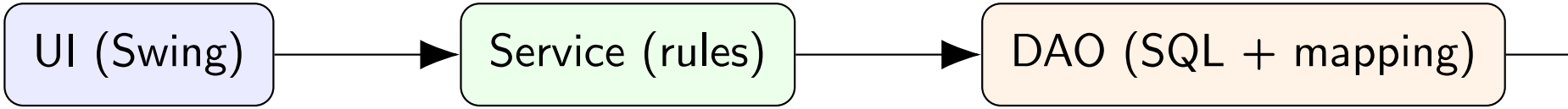

- UI: *no SQL*; convert fields + show results.
- Service: validations; business rules (e.g., no overlap).
- DAO: JDBC code only.

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Validation + user messages

- UI validation: required fields + parsing (numbers/dates).

Tip: never show stack traces to users.

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Validation + user messages

- UI validation: required fields + parsing (numbers/dates).
- Service validation: full rules (start before end; capacity > 0).
- Use `JOptionPane` for readable messages.

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EDT rule (avoid frozen UI)

- Swing is single-threaded (EDT paints + handles clicks).

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EDT rule (avoid frozen UI)

- Swing is single-threaded (EDT paints + handles clicks).
- Slow DB calls on EDT freeze the UI.
- If needed: use `SwingWorker` for background DB calls.

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Demo plan (10–15 mins)

1. Create `ResourceListPanel` with a `JTable`.

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Demo plan (10–15 mins)

1. Create `ResourceListPanel` with a `JTable`.
2. Add an "Add Resource" dialog (or panel).

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Demo plan (10–15 mins)

1. Create `ResourceListPanel` with a `JTable`.
2. Add an "Add Resource" dialog (or panel).
3. On save: call service → DAO insert.

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Demo plan (10–15 mins)

1. Create `ResourceListPanel` with a `JTable`.
2. Add an "Add Resource" dialog (or panel).
3. On save: call service → DAO insert.
4. Refresh table from DB.

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Mini Exercises (5 mins)

1. Which layer should enforce "booking end after start"?

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Mini Exercises (5 mins)

1. Which layer should enforce "booking end after start"?
2. If `Integer.parseInt` fails, where do you handle it?

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Mini Exercises (5 mins)

1. Which layer should enforce "booking end after start"?
2. If `Integer.parseInt` fails, where do you handle it?
3. What is the EDT and why does it matter?

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Key Takeaways

- Plan screens first; use layout managers for scalable UI.

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Key Takeaways

- Plan screens first; use layout managers for scalable UI.
- Keep button handlers short; call service methods.

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Key Takeaways

- Plan screens first; use layout managers for scalable UI.
- Keep button handlers short; call service methods.
- Deliverable-4 proves your app works end-to-end.

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Checkpoint Questions

1. Why should Swing UI avoid SQL code?

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Checkpoint Questions

1. Why should Swing UI avoid SQL code?
2. What is a good use of `CardLayout`?

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Checkpoint Questions

1. Why should Swing UI avoid SQL code?
2. What is a good use of `CardLayout`?
3. How do you refresh a table after inserting into DB?

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Next Lecture Preview

Next: Testing + build/artifacts + final demo (Deliverable-5)

- Add JUnit tests for service rules.

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Next Lecture Preview

Next: Testing + build/artifacts + final demo (Deliverable-5)

- Add JUnit tests for service rules.
- Package a runnable artifact and prepare a demo script.

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).